# SIMATS ENGINEERING
## ASSESSMENT TOOL 3

**COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN**
**COURSE CODE:CSA1104**

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**COURSE OUTCOME COVERED**

***CO2:*** Design UML diagrams to represent system requirements and architecture. *(BL3)*

*Assessment Tool Weightage: 30%*

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**QUESTIONS: 6** **Total Marks:30**

Annexure A

# Scenario-Based

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***Code of Conduct:***

*I _Dharshini/192521408_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***

*Dharshini*

Annexure A

# Scenario-Based

## 1. Use Case Diagram Scenario

A **Smart Cinema Ticket Booking System** allows customers to search movies, select seats, book tickets, cancel bookings, and make online payments. Theater managers update movie schedules and seat availability, while administrators manage users and system settings. The interactions between actors and system functions are not clearly defined.
Based on this scenario, analyze the system requirements, identify all actors and use cases, and design a **Use Case Diagram** showing appropriate relationships (**include, extend, and generalization**) to ensure complete system functionality.
**(5 Marks)**

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## 2. Class Diagram Scenario

A **Smart College Management System** consists of entities such as Student, Faculty, Department, Course, Attendance, and Examination. The existing design does not clearly define class attributes, operations, or relationships among these entities.
Based on this scenario, identify the required classes, define suitable attributes and methods, establish relationships (**association, aggregation, composition, and inheritance**), specify multiplicity, and construct a detailed **Class Diagram**.
**(5 Marks)**

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## 3. Sequence Diagram Scenario

In an **Online Railway Reservation System**, a passenger searches for trains, selects a seat, books a ticket, makes payment, and receives an e-ticket. The communication among the passenger, reservation system, payment gateway, and database is incomplete.
Based on this scenario, analyze the sequence of operations, identify participating objects and message flows, and design a **Sequence Diagram** representing the complete interaction process.
**(5 Marks)**

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## 4. Activity Diagram Scenario

A **Smart Insurance Claim Processing System** allows customers to submit insurance claims, upload supporting documents, verify claim details, approve or reject claims, and receive compensation. The workflow does not clearly represent decision-making or concurrent activities.
Based on this scenario, analyze the workflow, identify decision points and parallel activities, and design an optimized **Activity Diagram** using decision nodes, forks, and joins.
**(5 Marks)**

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## 5. State Diagram Scenario

A **Smart Washing Machine System** operates through states such as Idle, Water Filling, Washing, Rinsing, Spinning, Paused, and Completed. The transitions between these operational states are not clearly specified.
Based on this scenario, identify all possible states, triggering events, and transitions, and construct a **State Transition Diagram** that accurately represents the washing machine's behavior.
**(5 Marks)**

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## 6. Deployment Diagram Scenario

A **Smart Energy Management System** uses smart electricity meters, edge gateways, cloud servers, utility monitoring dashboards, and mobile applications to monitor energy consumption and control power distribution. The deployment architecture does not clearly illustrate hardware nodes, deployed software artifacts, or communication links.
Based on this scenario, analyze the deployment environment, identify all hardware nodes and software artifacts, and design a **Deployment Diagram** showing the physical architecture and communication links.
**(5 Marks)**

**Answer:**

## 1. Use Case Diagram – Smart Cinema Ticket Booking System

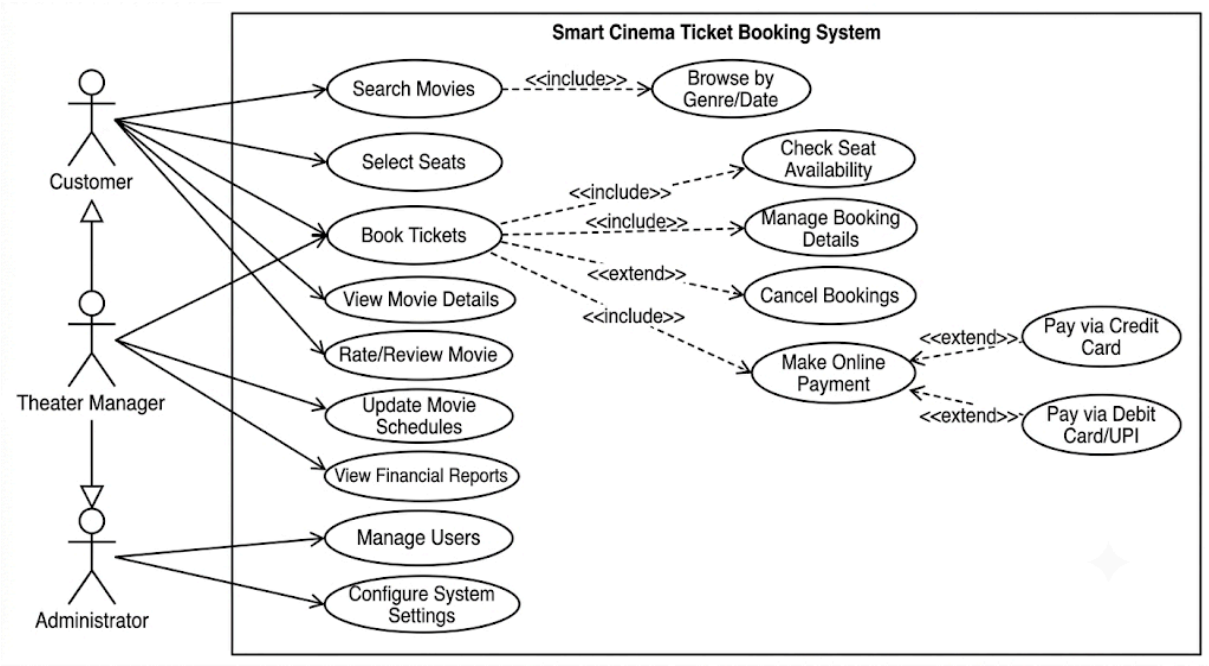
**Actors**

- Customer: Searches movies, selects seats, books or cancels tickets, and makes payments.
- Theater Manager: Updates movie schedules and seat availability.
- Administrator: Manages users and system settings.
- Payment Gateway: Processes online payments.

**Use cases**

Search Movies, Select Seats, Book Tickets, Make Payment, Cancel Booking, Update Movie Schedule, Manage Seat Availability, Manage Users, and Manage System Settings.

**UML Use Case Diagram**




The `include` relationship represents steps required for booking, such as seat selection, booking validation, and payment. The `extend` relationship represents an optional refund process when a booking is cancelled. The actors interact with the system according to their roles.

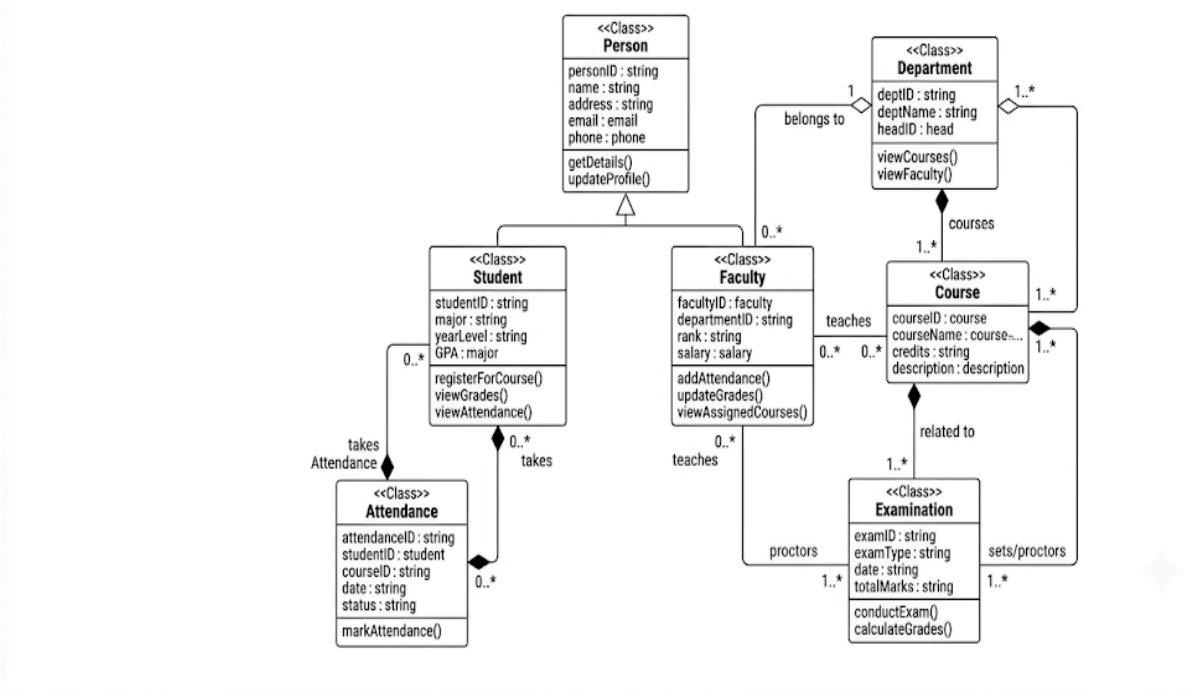
Note: This diagram uses a generalization-free design because the scenario does not require specialized actor types. Include and extend relationships are shown where they add meaningful functionality.

## 2. Class Diagram – Smart College Management System

**Classes and attributes**

- Student: studentId, name, email; methods: registerCourse(), viewAttendance().
- Faculty: facultyId, name, designation; methods: teachCourse(), markAttendance().
- Department: departmentId, departmentName; methods: addCourse(), assignFaculty().
- Course: courseId, courseName, credits; methods: enrollStudent(), viewCourseDetails().
- Attendance: attendanceId, date, status; methods: markPresent(), updateStatus().
- Examination: examId, examName, examDate; methods: scheduleExam(), publishResults().

**UML Class Diagram**



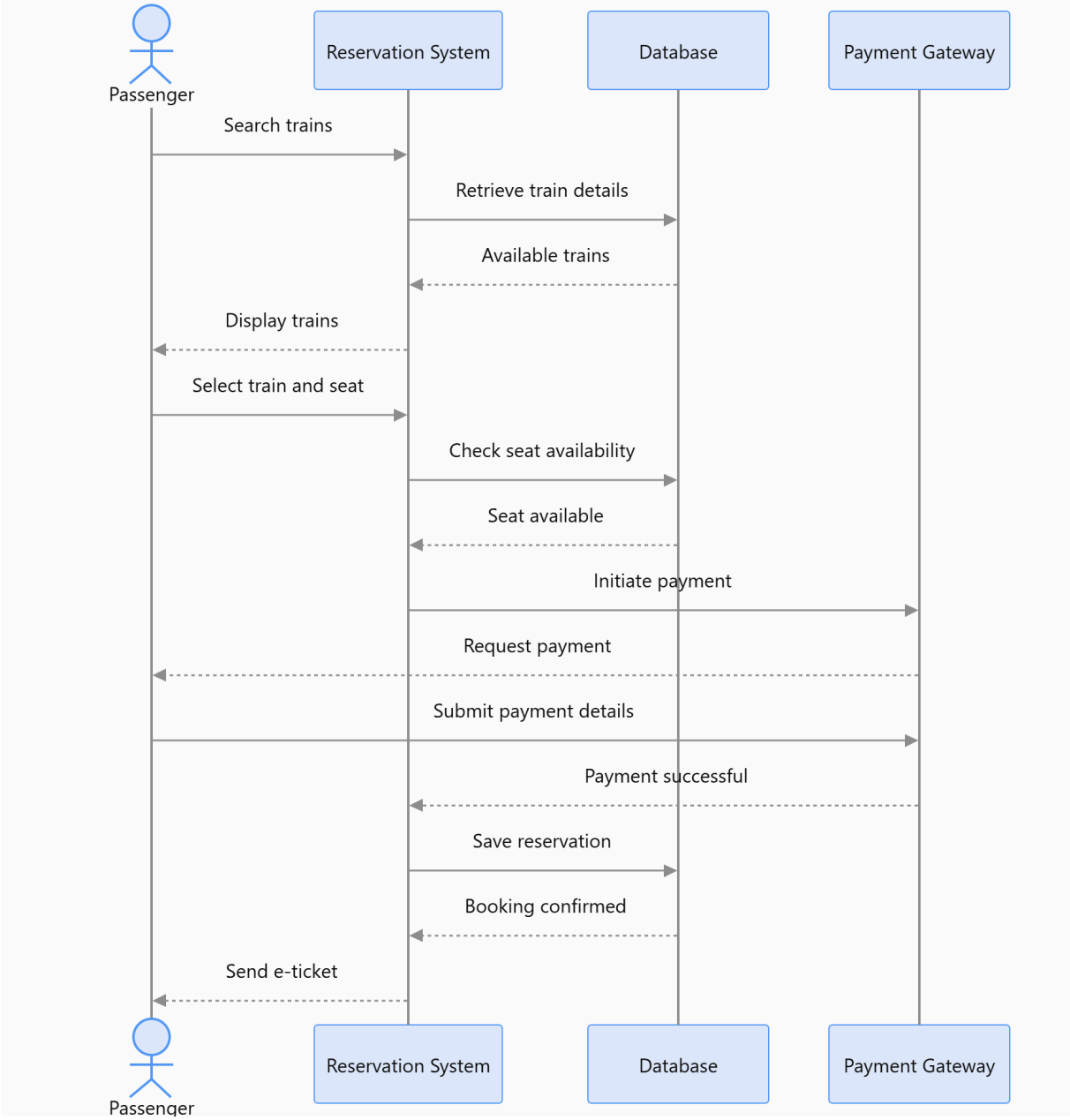

Inheritance is represented by `Person` as the parent class of `Student` and `Faculty`. Association connects students with courses. Aggregation shows that a department offers courses and employs faculty. Composition shows that course-related attendance records and examinations belong to a course in this model. Multiplicity specifies how many objects can participate in each relationship.

## 3. Sequence Diagram – Online Railway Reservation System

**Participating objects**

- Passenger: Searches for trains and books tickets.
- Reservation System: Handles train search, seat selection, and booking.
- Payment Gateway: Processes payment.
- Database: Stores train, seat, and reservation details.

**UML Sequence Diagram**




The passenger searches for trains and selects an available seat. The reservation system checks seat availability and initiates payment through the payment gateway. After successful payment, the reservation is saved in the database, and an e-ticket is sent to the passenger.

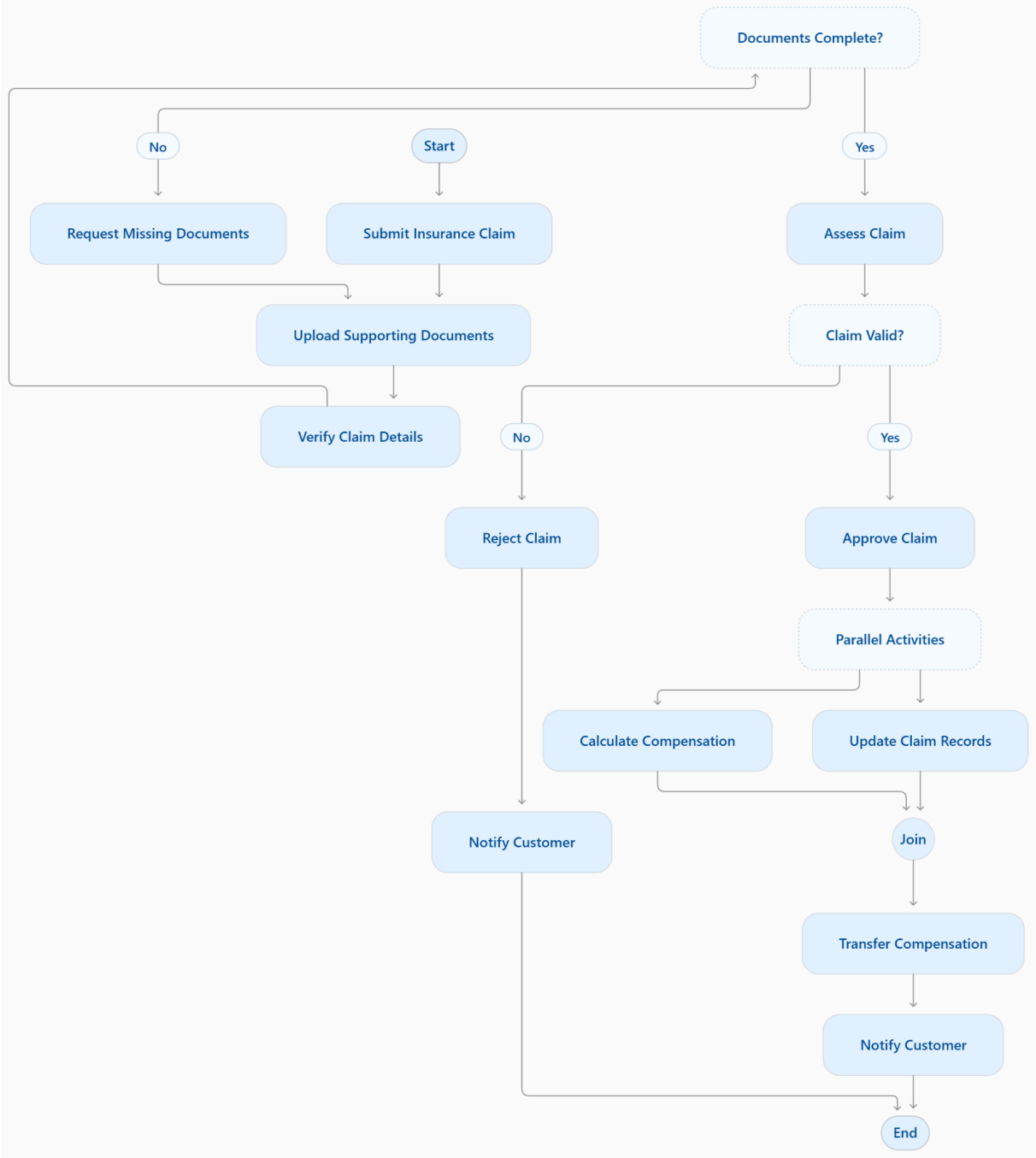
## 4. Activity Diagram – Smart Insurance Claim Processing System

**Workflow**

1. The customer submits an insurance claim.
2. The customer uploads supporting documents.
3. The system verifies the claim details.
4. The claim is checked for completeness and validity.

5. Valid claims proceed to assessment, while invalid claims are rejected or returned for correction.
6. Approved claims receive compensation.

**UML Activity Diagram**



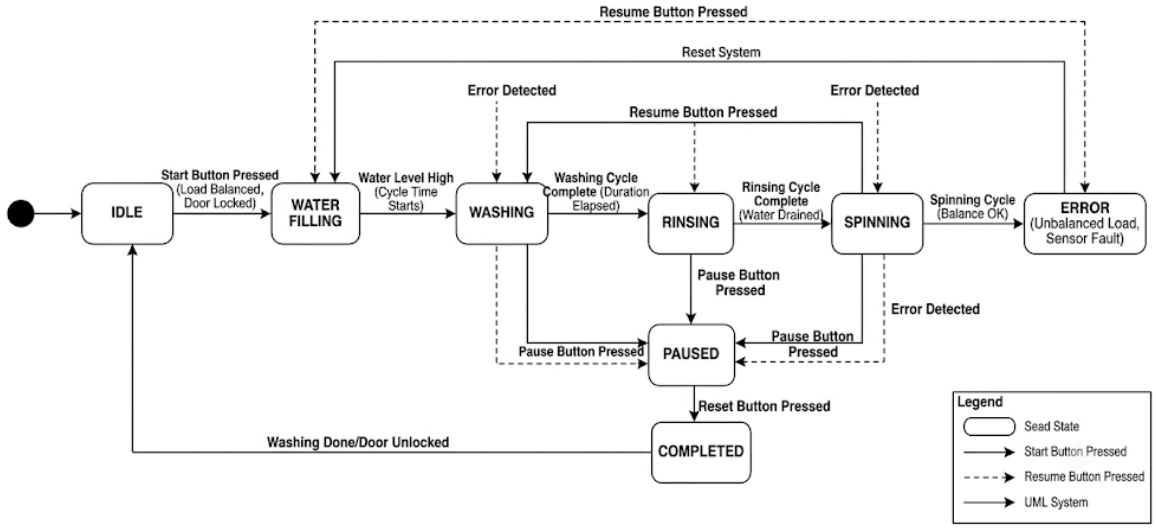

Decision nodes determine whether documents are complete and whether the claim is valid. Incomplete documents are requested again. Valid claims are approved, and compensation calculation and record updating occur concurrently. A join synchronizes both activities before compensation is transferred.

## 5. State Diagram – Smart Washing Machine System

**States and transitions**

- Idle: The machine waits for a user to start a wash cycle.
- Water Filling: Water enters the drum.
- Washing: The drum washes the clothes.
- Rinsing: Clothes are rinsed with clean water.
- Spinning: Excess water is removed.
- Paused: The current cycle is temporarily stopped.
- Completed: The washing cycle has finished.

**UML State Transition Diagram**



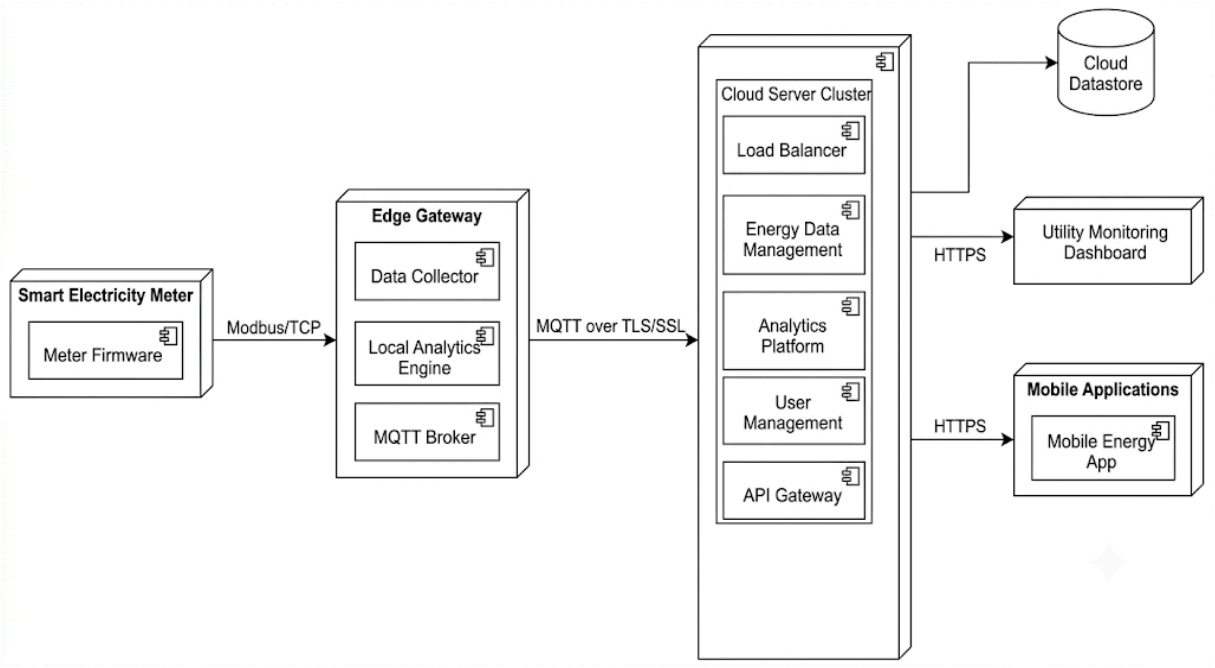

The machine moves through the washing states in sequence. Events such as `Start wash`, `Pause`, and `Resume` trigger transitions between states. After spinning finishes, the machine enters the Completed state. It can then return to Idle for another cycle.

## 6. Deployment Diagram – Smart Energy Management System

**Hardware nodes and software artifacts**

- Smart Electricity Meter: Measures electricity consumption and runs meter-monitoring software.
- Edge Gateway: Collects meter readings and forwards data to the cloud.
- Cloud Server: Runs energy analysis, monitoring, and power distribution control services.
- Utility Monitoring Dashboard: Allows utility staff to monitor consumption and control power distribution.
- Mobile Application: Allows users to view energy usage and receive alerts.

**UML Deployment Diagram**




The deployment diagram represents the physical arrangement of hardware nodes and the software deployed on them. Smart electricity meters send energy readings to edge gateways, which forward the information to cloud servers. The cloud stores and analyzes the readings in an energy database. Utility staff access the monitoring dashboard, while consumers use the mobile application to view energy consumption. Communication links such as Zigbee, MQTT, Wi-Fi, and HTTPS connect the components.

**RUBRICS FOR CALCULATION**

# Scenario-Based

| Criteria | Weightage | Level 4 (Exemplary) | Level 3 (Proficient) | Level 2 (Developing) | Level 1 (Beginning) |
|---|---|---|---|---|---|
| Understanding of Scenario | 20% | Demonstrates clear and complete understanding of the scenario and context | Shows good understanding with minor gaps | Partial understanding; misses key aspects | Misinterprets or fails to understand the scenario |
| Identification of Key Issues | 20% | Accurately identifies all relevant issues and factors involved | Identifies most key issues with minor omissions | Identifies limited issues; some irrelevant points | Unable to identify key issues |
| Application of Concepts | 25% | Applies appropriate concepts effectively to address the scenario | Applies concepts with minor errors | Limited or partially correct application | Unable to apply relevant concepts |
| Decision Making | 20% | Makes well-reasoned, logical decisions with strong rationale | Makes reasonable decisions with some justification | Makes weak decisions with limited justification | No clear decisions made or rationale provided |
| Clarity of Response | 15% | Response is clear, structured, and logically presented | Generally clear with minor issues | Somewhat unclear or poorly structured | Disorganized and difficult to understand |